

Smart Manufacturing Notes

S 3 & 11

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Chapter 11 · VR & AR

Exam Ready

CHAPTER 3

Five-Dimension Digital Twin Modeling and Its Key Technologies

From Grieves' 3D model to the extended 5D framework, three DT levels, and 8 modeling rules

3

3.1 — The Traditional Three-Dimension Digital Twin (Grieves, 2003)

ORIGIN

First proposed by Grieves in 2003 at the University of Michigan in a Product Lifecycle Management (PLM) course. Considered the origin of the DT concept.

The original three-dimension DT contains three parts:

PHYSICAL ENTITY

Exists in physical space. Has real functions and capabilities. Completes missions and produces actual outputs (e.g., aircraft, radar).

VIRTUAL ENTITY

A digital mirror of the PE. Described by models of geometry, physics, behavior. Characterized by real-time sync, faithful mapping, and high fidelity.

CONNECTION

Enables bidirectional data exchange. PE data → VE for model updating. VE insights → PE to guide and optimize. Forms a closed loop.

LIMITATION

The 3D model cannot accommodate new requirements from wider applications, new IT (IoT, cloud, AI), and the need for data fusion and on-demand services.

3.2 — New Requirements on the Digital Twin

1. WIDER APPLICATION

DT expanded beyond aerospace into automotive, healthcare, agriculture. New demands: high-quality service, user demand mining, shorter time-to-market.

2. EMBRACE NEW IT

IoT, cloud computing, big data, AI build a closed loop between physical and virtual spaces. The 3D DT couldn't fully leverage these technologies.

3. DATA & SERVICES

Data is crucial — physical + virtual sources must be integrated. Services must be encapsulated with friendly interfaces for non-expert users.

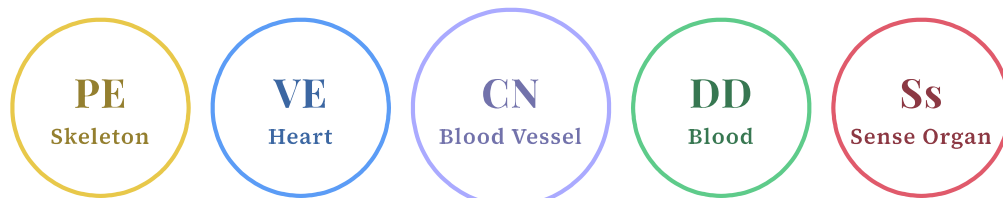
4. HIGH-FIDELITY MODELING

Need geometry + physics + behavior + rules models — all integrated. Behavior and rules modeling were underdeveloped in the 3D DT.

3.3 — The Extended Five-Dimension Digital Twin

$$M_{DT} = (PE, VE, Ss, DD, CN)$$

The DT is extended to 5 dimensions by adding DT Data (DD) and Services (Ss). Using a human body analogy:



PE — PHYSICAL ENTITY ("SKELETON")

Consists of subsystems (dynamic, control, hydraulic) + sensors. Sensors collect real-time states (vibration, force, torque, speed) and conditions (environment, loads, control orders). Requires RFID, WSN, embedded systems, soft sensors for data collection.

VE — VIRTUAL ENTITY ("HEART") — EQUATION 3.2: $VE = (GV, PV, BV, RV)$

A faithful mirror of PE consisting of four coupled model types:

Gv — Geometry Model

Describes shapes, sizes, assembly relations. The "container" for other models. Built with SolidWorks, AutoCAD, 3D MAX.

Pv — Physics Model

Physical properties (speed, wear, force) based on Gv. Reflects deformation, fracture, corrosion. FEM widely used.

Bv — Behavior Model

State transitions, performance degradation, coordination under control orders or disturbances. Uses FSM, Markov chain, ontology.

Rv — Rule Model

Rules (associations, constraints, deductions) mined from historical PE data. Gives VE prediction capability. Neural networks, SVM, random forests.

After modeling, Gv, Pv, Bv, Rv are coupled and tested via VV&A (Verification, Validation, Accreditation).

SS — SERVICES ("SENSE ORGAN") — EQUATION 3.3: $SS = (FUNCTION, INPUT, OUTPUT, QUALITY, STATE)$

Each service is a quintuple. For PE: monitoring, PHM, state prediction, energy optimization. For VE: construction, calibration, model test services. Enables on-demand usage by non-expert users.

DD — DIGITAL TWIN DATA ("BLOOD") — EQUATION 3.4: $DD = (DP, DV, DS, DK, DF)$

Dp: PE data (operation states, conditions) · Dv: VE data (model params, simulation outputs) · Ds: Service data (encapsulation, invocation) · Dk: Domain knowledge (mined or from databases) · Df: Fused data (via neural network, D-S algorithm, Kalman filter). DD is the *driver* of the DT.

CN — CONNECTION ("BLOOD VESSEL") — EQUATION 3.5

$CN = (CN_SD, CN_PD, CN_VD, CN_PS, CN_VS, CN_PV)$ — all **bidirectional**. Each connection modeled as a quintuple: (Data sources, Unit, Value, Scope, Sampling interval).

CN_SD	Ss ↔ DD	Services consume & produce data in DD
CN_PD	PE ↔ DD	Sensor data stored in DD; processed data returned to PE
CN_VD	VE ↔ DD	Simulation outputs stored; DD updates VE initialization
CN_PS	PE ↔ Ss	Real data drives services; service commands actuate PE
CN_VS	VE ↔ Ss	Services run on VE; validate strategies before applying to PE
CN_PV	PE ↔ VE	Real-time synchronization; used for calibration when PE/VE deviate

3.4 — Application-Oriented Three-Level Digital Twins

DTs are classified into three levels based on the complexity of the physical entity they model (using a shop-floor as example):

UNIT-LEVEL DT

Minimum independent individual (one person, one machine, one product). Contains a basic PE↔VE closed loop. Supports state perception, analysis, self-control. Applications: monitoring and PHM of a single machine.

SYSTEM-LEVEL DT

Multiple unit-level DTs connected. Characterized by self-organization, self-configuration, self-optimization. New services added: scheduling, progress control, process optimization for multiple units.

SOS-LEVEL DT

System-level DTs combined into a System of Systems. Global optimization across the entire shop-floor. Driven by data aggregation, distributed computing, model interoperation. Most complex — abundant services and data.

EXAM TIP

The three levels map to scale: unit = one machine, system = production line, SoS = entire shop-floor. Use this to justify which DT level is appropriate for a given application scenario.

3.5 — Key Technologies for Each Dimension

PE TECHNOLOGIES

- › RFID — track & identify PE
- › WSN — distributed sensors for state collection
- › Soft sensor — calculate hard-to-measure params via algorithms
- › Embedded system — for monitoring & control
- › Distributed sensor layout optimization

VE TECHNOLOGIES

- › 3D solid modeling (Gv)
- › Physics modeling — FEM, multiphysics (Pv)
- › Behavior modeling — FSM, Markov chain (Bv)
- › Rule modeling — ML algorithms (Rv)
- › Model integration & consistency analysis
- › VV&A (Verification, Validation, Accreditation)

SS TECHNOLOGIES

- › Resource management, description, encapsulation
- › Service visualization & HMI
- › Service search, matching, evaluation
- › Optimal selection & combination
- › Fault-tolerant management

DD TECHNOLOGIES

- › Data storage, modeling, cleaning
- › Data transformation & integration/fusion
- › Data analysis & visualization
- › Data mining — statistical / ML
- › Data security & integrity check

CN TECHNOLOGIES

- › Communication protocol analysis & conversion
- › Devices for perception and access (unified protocol conversion)
- › Wireless communication for flexible transmission
- › API design for software-to-software communication
- › Communication standards and specifications

3.6 — Eight Rules for Digital Twin Modeling

01

DATA & KNOWLEDGE BASED

Existing knowledge provides rules; real-time data continuously calibrates models.

02

MODULARIZATION

Separate & recombine components (unit/system/SoS) for flexibility and reusability.

03

LIGHT WEIGHT

Shorter transmission time, faster speed, better real-time performance. Simplify without losing fidelity.

04

HIERARCHY

Edge-fog-cloud structure distributes resources at appropriate layers for timeliness.

05

STANDARDIZATION

Unified encoding, interfaces, protocols — enables module integration & data exchange.

06

SERVITIZATION

Encapsulate DT functions (evaluation, prediction, validation) as standard services for easy use.

07

OPENNESS & SCALABILITY

Open to external models/services/tools; scalable through adding new models.

08

ROBUSTNESS

Handles uncertainty, dynamism, complexity of industrial environments. Adapts when conditions change.

Chapter 3 — Summary

- › Grieves' 3D DT (PE + VE + CN) is insufficient for modern needs
- › Extended 5D DT adds DD (data fusion driver) and Ss (on-demand services)
- › $VE = Gv + Pv + Bv + Rv$ (4 coupled model types); tested by VV&A
- › CN has exactly 6 bidirectional connections: CN_SD, CN_PD, CN_VD, CN_PS, CN_VS, CN_PV
- › Three DT levels: unit (one machine) → system (production line) → SoS (entire shop-floor)
- › 8 modeling rules guide future DT development

Digital Twin and VR / AR / Mixed Reality

Applications, comparison, integration framework, and assembly case study

11

11.1 — Introduction & Definitions

VR — VIRTUAL REALITY

Provides a fully virtual, immersive computer-generated environment. Users are isolated from the physical world. Characterized by immersion, presence, interactivity, navigation. Devices: HMDs, data gloves, CAVE systems.

AR — AUGMENTED REALITY

Overlays virtual elements on the real world — augments rather than replaces reality. Term coined by Caudell & Mizell (1990). User stays in physical environment supplemented by virtual info.

MR — MIXED REALITY

First proposed by Milgram & Kishino (1994). Extends along a "virtuality continuum" from fully real to fully virtual. Essentially the same as AR. In this chapter, AR is used to denote AR/MR.

WHY VR/AR + DT?

VR and AR can create interactive, immersive environments that **enhance the seamless integration of physical and virtual spaces**, which is exactly what the DT needs. Combining them augments the DT's physical-virtual fusion and improves service quality for users.

11.2 — VR in Design, Manufacturing, and Service

VR IN DESIGN

- › Factory layout design in 3D virtual space
- › Collaborative design platforms (multiple

VR IN MANUFACTURING

- › CNC machining simulation & tool path planning
- › Virtual assembly verification (collision

VR IN SERVICE

- › Maintenance simulation & visualization
- › Logistics planning (immersive warehouse)

users)

- › Product prototyping & design evaluation
- › Stress evaluation, interference checking
- › Cable harness design with captured knowledge

detection)

- › Multimodal interaction (gesture + speech)
- › Robot path planning via virtual arms
- › Haptic-based programming systems

- › Operator training for assembly sequences
- › Maintenance training with disassembly planning
- › Rocket engine health management visualization

VR LIMITATIONS

Users are isolated from the familiar physical environment, decreasing fidelity. Building a completely virtual environment imposes heavy computational burden. Real-time performance is challenging.

11.3 — AR in Design, Manufacturing, and Service

AR IN DESIGN

- › Overlay 3D machine models on physical space for factory layout
- › Gesture-based prototype evaluation
- › Collaborative product design across locations
- › Reformable mock-ups for shape/color testing
- › MR usability tests for household appliances

AR IN MANUFACTURING

- › In-situ CNC simulation (real cutter + virtual workpiece)
- › NC program validation with AR overlay
- › Assembly planning: WDP + PDP integration
- › Assembly guidance (visual tree + RFID)
- › Robot path planning preview before execution

AR IN SERVICE

- › Equipment maintenance with 3D model overlays
- › Voltage test guidance via AR markers
- › Warehouse logistics with RFID + AR visualization
- › Order picking with HMD (position + process info)
- › Assembly training with intelligent tutoring (ITS + AR)

11.4 — Comparison: VR vs AR

ASPECT	VR (VIRTUAL REALITY)	AR / MR (AUGMENTED REALITY)
ENVIRONMENT	Fully virtual — completely replaces physical world	Overlays virtual objects onto the real physical world
IMAGE SOURCE	Virtual graphics, images, models only	Combination of physical objects + virtual elements
PERSPECTIVE	Virtual-world centric; objects scale to virtual perspective	Physical-world centric; virtual objects align to real surroundings
PRESENCE	In the virtual world; no sense of physical world	In the physical world, augmented with virtual objects
INTERACTION	Only with virtual objects via HMDs, gloves, handles	With both physical and virtual objects simultaneously, more natural
IMMERSION	Full immersion; visual, auditory, haptic feedback in virtual world	Familiar physical environment + supplementary virtual elements; retains realism
EXAMPLE DEVICES	Oculus Rift, HTC Vive, Sony PlayStation VR, Samsung Gear VR	Microsoft HoloLens, Meta Headsets, Google Glass, ODG headsets
BEST FOR IN DT	Planning, training, verification in fully simulated environment	Guidance, maintenance, operation with physical context retained

11.5 — Digital Twin + VR and AR: Framework

Existing company applications demonstrate the trend:

VR IN DT (INDUSTRY)

- › Siemens: VR goggles for gas turbine DT; sensor data → colors for temp distribution
- › Huawei: DTs for products + assembly stations; VR glasses/gloves for virtual assembly simulation

AR IN DT (INDUSTRY)

- › GE: HoloLens overlays virtual steam turbine on real one — identifies damaged parts by "looking inside"
- › PTC: DT + AR/VR combo for machine maintenance guidance

- › **Oracle:** Users visit DT-based virtual plant via Gear VR
- › **Alibaba Cloud:** DT for integrated data model to improve existing VR systems

- › **Microsoft:** MR in process DT — holograms of production line with real-time data for remote collaboration
- › **Peaxy:** AR data visualization essential for DT to solve real business problems

APPLICATION FRAMEWORK — HOW VR AND AR WORK IN THE 5D DT

With VR: High-fidelity VEs presented in synthetic environment. User inputs (sounds, operations, positions) captured via VR devices → VE behaves autonomously → insights from DD demonstrated as visual/auditory/haptic feedback → supports layout planning, manufacturing operation, maintenance decisions.

APPLICATION FRAMEWORK — AR ROLE

With AR: Gestures, operations, positions of PE captured by cameras/gloves/RFIDs → through image processing, tracking, rendering → virtual models from VE and insights from DD superimposed on PEs → displayed via HMDs, iPads, projectors → mixed environment where PE + VE + DD are tightly coupled → more natural interaction in design, operation, maintenance.

11.6 — Case Study: DT-Driven Assembly Combining VR and AR

Using a **complex aerospace product assembly** as example. Key physical elements:

- › **Articulated arm + IPC:** Scans parts to measure geometric dimensions and tolerances → point cloud data for virtual modeling
- › **Laser tracker + IPC:** Tracks parts in real-time, measures 3D coordinates and key features → assembly accuracy prediction and process control
- › **Laser projector + IPC:** Projects CAD-based outlines onto physical space to guide assembly and precise positioning
- › **VR/AR devices + IPC:** HMDs, gloves, handles capture physical scene data; IPC performs simulation, tracking, information rendering
- › **Virtual models, data and services:** Assembly design model, process model, demonstration model, scene model — all built from physical data; fused with real data to calibrate and optimize

VR ROLE IN ASSEMBLY

VEs operate in immersive artificial environment
→ high-quality assembly planning, verification,
operator training *before* actual assembly →
eliminates costly errors in practice. Iterative
design with no physical cost.

AR ROLE IN ASSEMBLY

Overlays graphical elements and DD insights on
PEs → just-in-time information for assembly
guidance → natural interaction (voice + gesture) →
operator stays in physical environment while
receiving virtual guidance.

Chapter 11 — Summary

- › VR = fully virtual, immersive; isolates user from physical world; best for planning, training, design verification
- › AR/MR = overlays virtual on real; user stays in physical environment; best for real-time guidance, maintenance, operation
- › VR is limited by isolation from reality and heavy computational cost; AR retains physical context but needs tracking accuracy
- › In the 5D DT framework: VR enhances VE+DD interaction; AR integrates PE+VE+DD into a mixed environment
- › Industry leaders (GE, Siemens, PTC, Microsoft) are already combining DT with VR/AR
- › Assembly case: articulated arm + laser tracker + laser projector + VR/AR devices all feed the 5D DT to optimize aerospace assembly
- › Combining VR/AR with DT is described as "*an inevitable and beneficial trend*"

KEY EXAM POINTS

Q: Define VR and AR → VR = fully virtual immersive environment; AR = virtual information overlaid on real world. Q: How does VR/AR fit into 5D DT? → VR works with VE+DD in synthetic space; AR merges PE+VE+DD in physical space. Q: Give one industrial example each → VR: Siemens gas turbine DT with VR goggles; AR: GE HoloLens steam turbine inspection.